

Form Validation Manual Method

project structure



index.html

```
<!DOCTYPE html>
<html lang="en">
<head>
  <meta charset="UTF-8">
  <meta name="viewport" content="width=device-width, initial-scale=1.0">
  <title>Home Page</title>
</head>
<body>
  <a href="/signup">Signup</a>
  <a href="/login">Login</a>
  <h1>Welcome to the Application</h1>
</body>
</html>
```

login.html

```
<!DOCTYPE html>
<html lang="en">
<head>
  <meta charset="UTF-8">
  <meta name="viewport" content="width=device-width, initial-scale=1.0">
  <title>Application Login</title>
</head>
<body>
  <h2>Application Login</h2>
  <form action="/login" method="post">
    <label for="username">Username:</label><br/>
```

```

        <input type="text" id="username" name="username" value="{{
form_data.get('username', '') }}" required><br/>

        <label for="password">Password:</label><br/>
        <input type="password" id="password" name="password" required><br/>

        <button type="submit">Login</button>
    </form>
    <p>Don't have an account? <a href="/signup">Signup here</a></p>

    {% if errors %}
        <ul style="color: red;">
            {% for err in errors %}
                <li>{{ err }}</li>
            {% endfor %}
        </ul>
    {% endif %}
</body>
</html>

```

signup.html

```

<!DOCTYPE html>
<html lang="en">
<head>
    <meta charset="UTF-8">
    <meta name="viewport" content="width=device-width, initial-scale=1.0">
    <title>Signup Page</title>
</head>
<body>
    <h2>Application Signup</h2>
    <form action="/signup" method="post">
        <label for="first_name">First Name:</label><br/>
        <input type="text" id="first_name" name="first_name" value="{{
form_data.get('first_name', '') }}" required><br/>
        <label for="last_name">Last Name:</label><br/>
        <input type="text" id="last_name" name="last_name" value="{{
form_data.get('last_name', '') }}" required><br/>
        <label for="username">Username:</label><br/>
        <input type="text" id="username" name="username" value="{{
form_data.get('username', '') }}" required><br/>
        <label for="password">Password:</label><br/>
        <input type="password" id="password" name="password" required><br/>
        <button type="submit">Signup</button>
    </form>

```

```

<p>Already have an account? <a href="/login">Login here</a></p>

{% if errors %}
    <ul style="color: red;">
        {% for err in errors %}
            <li>{{ err }}</li>
        {% endfor %}
    </ul>
{% endif %}

</body>
</html>

```

app.py

```

from flask import Flask, render_template, request
import re

app = Flask(__name__)
@app.route('/')
def index():
    return render_template('index.html')

@app.route('/signup', methods=['GET', 'POST'])
def signup():
    if request.method == 'POST':
        # capture form data
        form_data = request.form
        first_name = request.form['first_name']
        last_name = request.form['last_name']
        username = request.form['username']
        password = request.form['password']
        print(first_name, last_name, username, password)

        #form validation
        errors = []
        if not first_name:
            errors.append("First name is required.")
        if not first_name or not re.match(r'^[A-Za-z]', first_name):
            errors.append("First name must start with an alphabet.")
        if not last_name:
            errors.append("Last name is required.")
        if not last_name or not re.match(r'^[A-Za-z]', last_name):
            errors.append("Last name must start with an alphabet.")
        if not username:
            errors.append("Username is required.")

```

```

        if not username or not re.match(r'^[A-Za-z][A-Za-z0-9]*$', username):
            errors.append("Username must start with a letter and contain only
letters and numbers.")

        if not password or not is_strong_password(password):
            errors.append("Password must be at least 6 characters long and
include a capital letter, lowercase letter, number, and special character.")
        if errors:
            return render_template('signup.html', errors=errors,
form_data=form_data)
        pass
        return render_template('signup.html', form_data={})

def is_strong_password(password):
    if (len(password) < 6 or
        not re.search(r'[A-Z]', password) or
        not re.search(r'[a-z]', password) or
        not re.search(r'\d', password) or
        not re.search(r'[@#$$%^&*(),.?":{}|<>]', password)):
        return False
    return True

@app.route('/login', methods=['GET', 'POST'])
def login():
    if request.method == 'POST':
        # capture form data
        form_data = request.form
        username = request.form['username']
        password = request.form['password']
        print(username, password)
        #form validation
        errors = []
        if not username:
            errors.append("Username is required.")
        if not username or not re.match(r'^[A-Za-z][A-Za-z0-9]*$', username):
            errors.append("Username must start with a letter and contain only
letters and numbers.")
        if not password:
            errors.append("Password is required.")
        if errors:
            return render_template('login.html', errors=errors,
form_data=form_data)
        pass

```

```
        return render_template('login.html', form_data={})

if __name__ == '__main__':

    app.run(debug=True)
```